

Model Paper - Database Management Systems

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Q1 (20 marks)

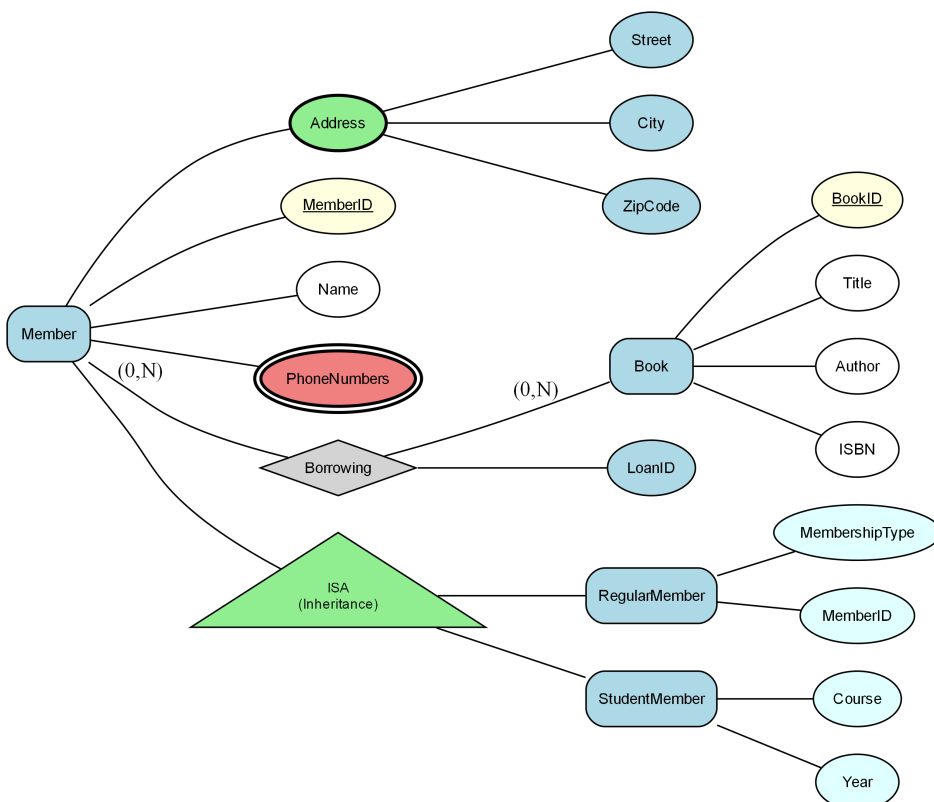
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

Consider a library management system. The system contains the following entities: Book (BookID, Title, Author, ISBN), Member (MemberID, Name, Address, PhoneNumbers), Loan (LoanID, LoanDate, ReturnDate), and Staff (StaffID, Name, Position). There are relationships among these entities such as Members borrowing Books with a Loan, where The Book entity has attributes such as BookID, Title, Author. has a relationship with the Author, where multiple authors can contribute to a book. The Member entity has attributes such as MemberID, Name, Address. can be specialized into RegularMember (with attributes MembershipType) and StudentMember (with attributes Course, Year). A Member can be related to many Books, and a Book can be related to many Members through the 'Borrowing' relationship (Member participation is optional, Book participation is optional)

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{ \{AB \rightarrow C, B \rightarrow D, CD \rightarrow E, E \rightarrow A, AC \rightarrow B\} \}$

- a) Find all the keys in relation R using the attribute closure method. (4 marks)
- b) Is R in 3NF? Give reasons for your conclusion. (3 marks)
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A hospital has established a relational database to manage patient records, treatments, and billing information. The database consists of multiple tables, including Patients, Treatments, and Billing. Each table is designed to efficiently store and retrieve necessary information while adhering to relational database principles. The database administrator has been tasked with updating the structure of the Patients table to include additional fields, and they need to manipulate data entries across various tables based on new requirements.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)
- b) Code Segment: ````java String sql = "SELECT * FROM employees WHERE department = ?";
PreparedStatement pstmt = connection.prepareStatement(sql); pstmt.setString(1, "IT"); ResultSet rs =
pstmt.executeQuery(); ```` Which type of JDBC statement is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)
- c) Briefly explain the options available for modifying the existing structure of a table in SQL. (2 marks)
- d) Briefly explain how you would handle referential integrity in the billing and treatments tables related to the patients table. (3 marks)
- e) A hospital is establishing role-based access control for its database system that manages Patients, Doctors, and Appointments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Appointments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating patient appointment data.**
- Write a T-SQL statement to create a login for Sarah with Windows authentication. (2 marks)
 - Provide Sarah with the necessary permissions to handle all administrative tasks within the server using a fixed server role. (3 marks)
 - Assuming Emily's username is 'emily.e', write T-SQL statements to assign her the responsibility of managing the Appointments database using a user-defined server role. (5 marks)

iv. Assuming Nathan's login name is 'nathan.n', write T-SQL statements to grant him permission to create tables and databases. (3 marks)

v. Assuming Michael's login name is 'michael.m', write T-SQL statements to allow him to perform data entry tasks on the Patients and Appointments databases. (3 marks)

Q4 (40 marks)

Consider the following schema of a database designed for a healthcare facility: Patient (patientId: int, name: varchar(100), dob: date, address: varchar(150)) Appointment (appointmentId: int, patientId: int, appointmentDate: date, status: varchar(50), totalFee: real) Doctor (doctorId: int, name: varchar(100), specialty: varchar(100), contact: varchar(15)) Hospital (hospitalId: int, name: varchar(100), location: varchar(150), capacity: int). The 'Patient' table stores information about patients, 'Appointment' keeps track of appointments made by patients, 'Doctor' details registered doctors, and 'Hospital' contains information on hospitals. The 'Appointment' table holds appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Doctor' table manages details about doctors, including unique doctor ID, name, specialization, and contact information. The 'Hospital' table contains relevant information for the database system. Fee (feeId: int, patientId: int, amount: real, paymentStatus: varchar(20)) The 'Fee' table stores fee/payment details including payable amount and payment status.

a) Write SQL Queries to perform the following: i. Find the names and phone numbers of doctors who are specialists in 'Pediatrics'.

- i. Find the names and phone numbers of doctors who are specialists in 'Pediatrics'. (4 marks)
- ii. Find the patient with the highest aggregate value using GROUP BY on related appointment records. (6 marks)
- iii. Find the appointmentDate, status, and name from Appointment, Patient, and Appointment where Appointment.appointmentDate is not null. (7 marks)

b) b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)